

Part 1 - Data Warehouse Modeling: Star Schema

1.1 Define the Business Process and the Fact Grain

Business Process

The business process analyzed in this project is the execution of financial trading transactions recorded in the account statement dataset. These transactions include buying and selling activities associated with different stock symbols, industries, sectors, and geographical regions.

Fact Grain

The grain of the fact table is defined as: One row in Fact_Transactions represents one financial transaction from the account statement dataset.

Each record stores the transaction quantity together with references to the related dimensions (time, symbol, geography, and transaction type).

1.2 Identify Fact and Dimensions

Fact Table

Fact_Transactions

Measure

- quantity

Foreign Keys

- time_key
- symbol_key
- geo_key
- type_key

Dimension Tables

Dim_Time

time_key (PK)

date

day

month
quarter
year
day_of_week

Dim_Symbol

symbol_key (PK)
symbol
company
sector
industry

Dim_Geography

geo_key (PK)
country
country_code
region
sub_region

Dim_TransactionType

type_key (PK)
transaction_type

1.3 Define Dimension Hierarchies

Dim_Time Hierarchy

Day → Month → Quarter → Year

Dim_Geography Hierarchy

Country → Sub-region → Region

Dim_Symbol Hierarchy

Symbol → Industry → Sector

Dim_TransactionType Hierarchy

No hierarchy (single-level dimension)

1.4 Design the Star Schema

Surrogate Keys

Each dimension uses an integer surrogate key:

Dimension	Surrogate Key
Dim_Time	time_key
Dim_Symbol	symbol_key
Dim_Geography	geo_key
Dim_TransactionType	type_key

Fact Table Foreign Keys

Fact Table Column	References
time_key	Dim_Time
symbol_key	Dim_Symbol
geo_key	Dim_Geography
type_key	Dim_TransactionType

Measure

The quantitative measure stored in the fact table is:

quantity



Project Report: Financial Transactions Data Warehouse

1. Modeling Choices

For this project, I designed a **Star Schema** data warehouse model to support analytical processing and reporting of financial transactions. The model separates transaction measures from descriptive attributes, making analytical queries more efficient and simplifying dashboard development.

Fact Table (Fact_Transactions)

The central table of the schema is **Fact_Transactions**. Each row represents a single financial transaction from the account statement dataset.

The fact table contains:

- **quantity** (measure)
- **time_key** (FK)
- **symbol_key** (FK)
- **geo_key** (FK)
- **type_key** (FK)

Dimension Tables

Four dimensions were created:

Dim_Time

Contains temporal attributes used for time-based analysis:

- date
- day
- month
- quarter
- year
- day_of_week

Dim_Symbol

Contains company and stock-related information:

- symbol
- company
- sector
- industry

Dim_Geography

Contains geographical information:

- country
- country_code
- region
- sub_region

Dim_TransactionType

Contains transaction categories:

- BUY
- SELL
- DIVIDENT

All dimensions use surrogate integer keys to improve join efficiency and support dimensional modeling best practices.

2. Data Quality Issues Found

Several data quality checks were performed before building the final Star Schema.

Redundant Column

The transactions dataset contained an unnecessary column named **Unnamed: 5**.

The column contained only missing values and was removed during the cleaning phase.

Missing Values

Missing value analysis showed:

- 464 empty rows in the transaction dataset.
- The **Unnamed: 5** column contained only null values.

After removing incomplete rows and unnecessary attributes, the remaining transaction data contained no missing values in the required fields:

- Date
- TransactionType
- Symbol
- Unit

Referential Integrity Validation

A validation check was performed to ensure that every transaction symbol existed in the symbols reference dataset.

The process identified:

- **18 unmapped symbols**

These records could not be linked to the Symbol dimension and were excluded during the integration process through inner joins.

Country Mapping Validation

A second validation check was performed between the symbols dataset and the country reference dataset.

The process identified:

- **2 unmapped countries**

These records could not be linked to the Geography dimension and were excluded during the integration process.

Attribute Selection

Only attributes required by the dimensional model were retained.

Unused attributes were removed to keep the schema compact and consistent with the business requirements.

3. Main Analytical Results

After completing the ETL process and building the star schema, the final fact table contained:

842 valid transaction records

The analytical queries produced the following results.

Industry Activity in Q4 2024 (BUY Transactions)

The most active industries based on transaction count were:

Industry	Transactions
Semiconductors	16
Credit Services	7
Software - Infrastructure	7
Internet Content & Information	6
Telecom Services	6

Semiconductors recorded the highest number of BUY transactions during Q4 2024.

Total Units Bought by Region

Region	Total Units Bought
Americas	16,197
Europe	7,793
Asia	4,013

The Americas generated the highest buying volume among all regions.

Top Industries by Total Units Traded

Industry	Total Units Traded
Auto Manufacturers	1,869
Software - Application	951
Credit Services	932
Aerospace & Defense	887
Internet Content & Information	827

Auto Manufacturers recorded the highest trading volume across all industries.

Monday Selling Activity by Sector

Sector	Total Units Sold
Technology	975
Communication Services	659
Consumer Cyclical	506

The Technology sector generated the highest selling volume on Mondays.

Industrial Diversity by Region

Region	Distinct Industries
Americas	36
Europe	25
Asia	9

The Americas showed the highest industrial diversity, containing the largest number of distinct industries represented in the dataset.

Conclusion

The implemented Star Schema successfully transformed the operational datasets into an analytical data warehouse structure. The dimensional model supports efficient querying, enables multidimensional analysis across time, geography, symbols, and transaction types, and provides a solid foundation for interactive reporting through the Streamlit dashboard.